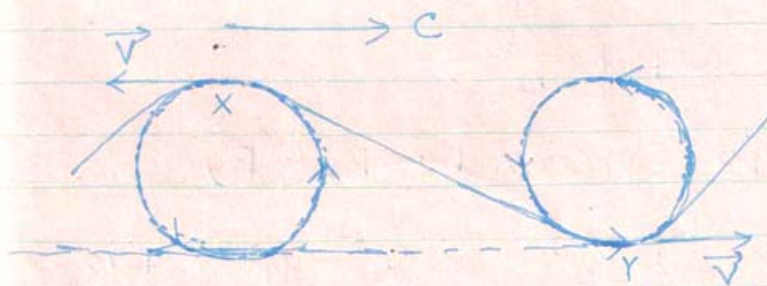
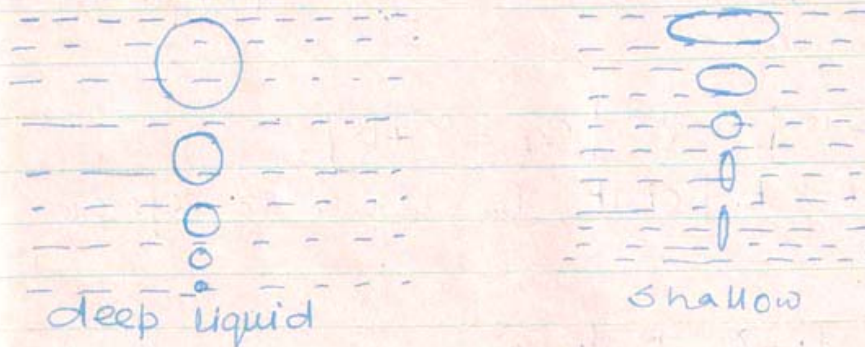




Ripples and Gravity Waves

Ripples and Gravity waves:

When a wave propagates through a deep liquid, the particles of the liquid describe circular motion. The radius of the circle decreases as the depth increases.



Let us consider a wave propagating through a liquid.

$\vec{c}$ = the velocity of propagation of the wave

λ = wave length of the wave.

r = Radius of the circle; described by the particles of the liquid.

ω = angular velocity of rotation of the particles.

V = linear velocity of the liquid particles describing circular motion = $r\omega$

The motion appears to be stable to an observer moving with the velocity $\vec{c}$ along the direction of propagation of the wave.



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But to an Observer at rest the actual motion is the resultant of a steady velocity $\vec{c}$ & Constant Speed v .

Let us consider two liquid particles at X & Y, the highest and the lowest point of the circle respectively.

Let q_1 and q_2 be the velocity at points X & Y respectively.

$$q_1 = c - v \quad \& \quad q_2 = c + v$$

Let the lowest point, be taken as reference level.

$$E_x = g \cdot 2r + \frac{1}{2} \cdot q_1^2$$

$$E_y = 0 + \frac{1}{2} q_2^2$$

Since energy is conserved: $E_x = E_y$

$$g \cdot 2r + \frac{1}{2} q_1^2 = \frac{1}{2} q_2^2 \quad \text{or} \quad q_2^2 - q_1^2 = 4gr$$

$$(q_2 + q_1)(q_2 - q_1) = 4gr$$

$$(c + v + c - v)(c + v - c + v) = 4gr$$

$$2c \cdot 2v = 4gr \quad \text{or} \quad c \omega r = gr$$

$$\text{or } c \cdot \frac{2\pi c}{\lambda} = g \quad \left[\because \omega = 2\pi n; n = \frac{c}{\lambda} \right]$$

$$\text{or } c^2 = \frac{g\lambda}{2\pi} \quad \text{or } c = \sqrt{\frac{g\lambda}{2\pi}} \quad \text{--- (1)}$$

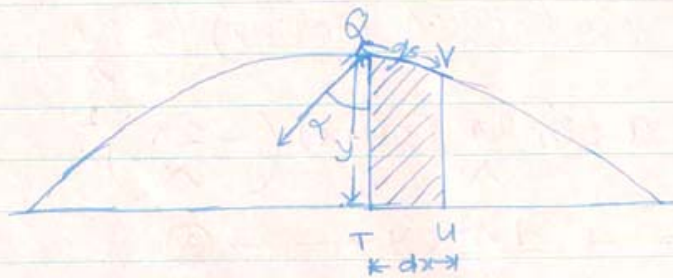
Equation (1) gives the velocity of propagation of a wave through a liquid, under the effect of gravity alone.



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We now consider; the effect of surface tension on the velocity of propagation of wave through a liquid.

Let us consider a plane wave, propagating through a liquid. In the fig. a section of the plane wave by a vertical plane is shown



Given: σ = Surface tension of the liquid.
 ρ = density of the liquid.

Let us consider an element of the wave $QTUV$ of height 'y' width 'dx' and actual length dx . Let 'r' & R be the principal radii of curvature at the point Q.

Excess pressure at Q: $P = \sigma \left(-\frac{1}{r} + \frac{1}{R} \right)$

Since for a plane wave; one of the principal radii of curvature 'R' is infinity ($R = \infty$)

$$\frac{1}{R} = 0$$

Excess pressure at Q; $P = \frac{\sigma}{r} \rightarrow \text{---}$

Excess pressure = $P = \frac{\sigma}{r} = P_{\text{concave side}} - P_{\text{convex side}}$

Using the relation from Calculus:-

$$\frac{1}{r} = \frac{d^2 y}{dx^2} \therefore P = \sigma \cdot \frac{d^2 y}{dx^2} \rightarrow (1)$$



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We know that the equation of a plane progressive wave is

$$y = a \sin \frac{2\pi}{\lambda} (ct - x) \rightarrow (2)$$

Differentiating Equation (2) w.r to x :-

$$\frac{dy}{dx} = a \cos \frac{2\pi}{\lambda} (ct - x) \left(-\frac{2\pi}{\lambda} \right)$$

$$\text{or } \frac{d^2y}{dx^2} = -a \sin \frac{2\pi}{\lambda} (ct - x) \left(-\frac{2\pi}{\lambda} \right)^2$$

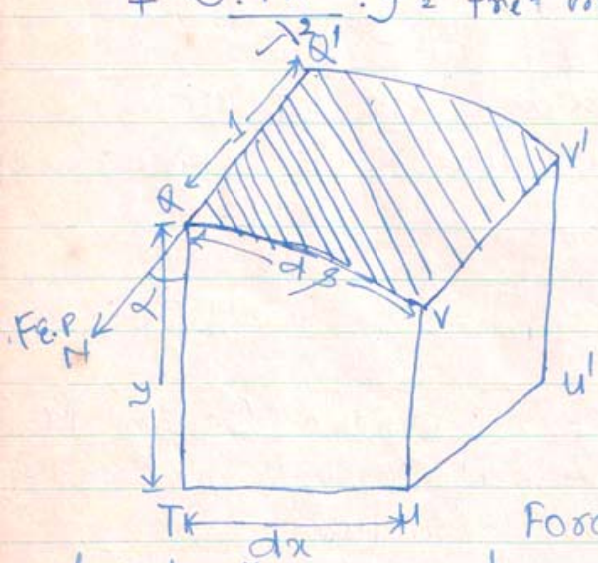
$$\text{or } \frac{d^2y}{dx^2} = -\frac{4\pi^2}{\lambda^2} \cdot y \rightarrow (3)$$

Putting Equation (3) in (1a)

$$\text{Excess pressure } P = \rho \left(-\frac{4\pi^2}{\lambda^2} y \right) = \text{Pre}^{\text{r}} \text{ on Concave Side} - \text{Pre}^{\text{r}} \text{ on Convex side}$$

The (-)ve sign indicates that the pressure on the Convex side is greater than the pressure on Concave side.

$$+ \frac{5.4\pi^2}{\lambda^2} \cdot y = \text{Pre}^{\text{r}} \text{ on the Convex} - \text{Pre}^{\text{r}} \text{ on the Concave} \rightarrow (4)$$



Let us consider an element of the wave of height y , breadth dx , width 'unit'

$$\text{Area of the curved surface} = (dx \times 1)$$

Force on the liquid element due to the excess pressure is along the normal $\vec{N}$



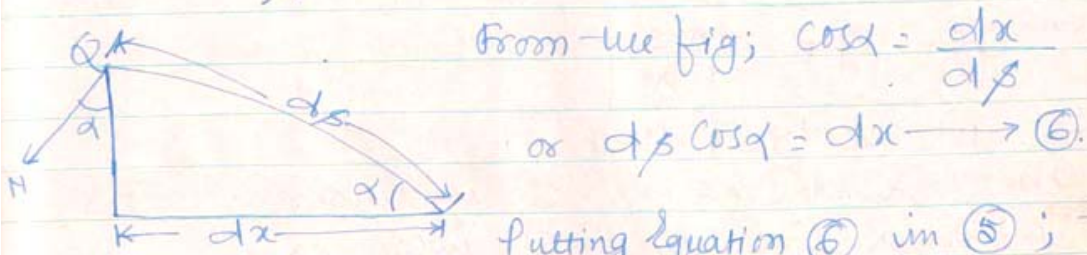
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to the curved surface = $p \times (d\phi \times 1) = \frac{4\pi^2 s y d\phi}{\lambda^2}$.

Let the normal $\vec{N}$ makes an angle α with the vertical.

The component of the force due to the excess pressure is the force due to the S.T along vertically downward direction.

$$= \frac{4\pi^2}{\lambda^2} s y d\phi \cos\alpha \longrightarrow (5)$$



From the fig; $\cos\alpha = \frac{dx}{d\phi}$

$$\text{or } d\phi \cos\alpha = dx \longrightarrow (6)$$

Putting Equation (6) in (5); the force on the liquid element; due to S.T along vertically downward direction

$$= \frac{4\pi^2}{\lambda^2} s y dx \longrightarrow (7)$$

The force on the liquid element; due to gravity along vertically downward direction side.

$$= (y \times dx \times 1) \rho g \longrightarrow (8)$$

Hence the total force on the liquid element along vertically downward direction; due to gravity as well as S.T

$$\begin{aligned} &= y \rho g dx + \frac{4\pi^2}{\lambda^2} s y dx \\ &= y \rho dx \left[g + \frac{4\pi^2 s}{\lambda^2 \rho} \right] \longrightarrow (9) \\ &= y \rho dx g' \end{aligned}$$

$$\text{where } g' = g + \frac{4\pi^2 s}{\rho \lambda^2} \longrightarrow (10)$$



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Thus from (10) we can conclude that; if we consider the effect of S.T along with gravity 'g' modifies to g' hence the velocity of propagation of wave through a liquid under the effect of gravity and S.T both; can be obtained on replacing 'g' in equation (1) by ' g' '

$$\text{i.e. } c = \sqrt{\frac{g' \lambda}{2\pi}} = \sqrt{\frac{g \lambda}{2\pi} + \frac{2\pi s}{\rho \lambda}} \rightarrow$$

$$\therefore g' = g + \frac{4\pi^2 s}{\rho \lambda^2}$$

From (11) we find that

When λ decreases, c increases.

When λ increases, c increases.

Hence there must be a particular value of ' λ ' for which ' c ' is minimum, that value of λ is known as λ_m . which can be found using the results from Calculus $\frac{dc}{d\lambda} = 0$

$$\frac{dc}{d\lambda} = \frac{1}{2} \left[\frac{g \lambda}{2\pi} + \frac{2\pi s}{\rho \lambda} \right]^{-\frac{1}{2}} \frac{d}{d\lambda} \left[\frac{g \lambda}{2\pi} + \frac{2\pi s}{\rho} \left(-\frac{1}{\lambda^2} \right) \right]$$

g, λ, s, ρ, π all being +ve; the 1st bracketed term being sum of two +ve quantities cannot be zero.

$$\frac{g}{2\pi} - \frac{2\pi s}{\rho \lambda^2} = 0$$

$$\frac{2\pi s}{\rho \lambda_m^2} = \frac{g}{2\pi}$$

$$\lambda_m^2 = \frac{4\pi^2 s}{\rho g}$$

$$\lambda_m = 2\pi \sqrt{\frac{s}{\rho g}} = \text{Const} \rightarrow (12)$$



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Thus for a given liquid; there is a particular Const. wave length given by equation (12) for which the velocity of the wave is minimum.

Case I: If the wave which propagates through the liquid has wave length λ greater than this value λ_m i.e. $\lambda > \lambda_m$ from (11) $\frac{g\lambda}{2\pi} > \frac{2\pi s}{\rho\lambda}$

Hence the effect of gravity is more important than the effect of surface tension and the wave is known as Gravity wave.

Case II: $\lambda < \lambda_m$ $\frac{2\pi s}{\rho\lambda} > \frac{g\lambda}{2\pi}$, the effect of

S.T is more important than the effect of gravity and the wave is known as Ripple.

$$C = \sqrt{\frac{g\lambda}{2\pi} + \frac{2\pi s}{\rho\lambda}} \quad \boxed{C = n\lambda}$$

$$n^2\lambda^2 = \frac{g\lambda}{2\pi} + \frac{2\pi s}{\rho\lambda}$$

$$\text{or } \frac{2\pi s}{\rho\lambda} = n^2\lambda^2 - \frac{g\lambda}{2\pi}$$

$$s = \frac{n^2\lambda^3\rho}{2\pi} - \frac{g\rho\lambda^2}{4\pi^2} \quad \text{--- (13)}$$

Using Equation (13), surface tension of the liquid can be experimentally determined.

Discussions: $W = mg$ or $g = \frac{W}{m}$ = the

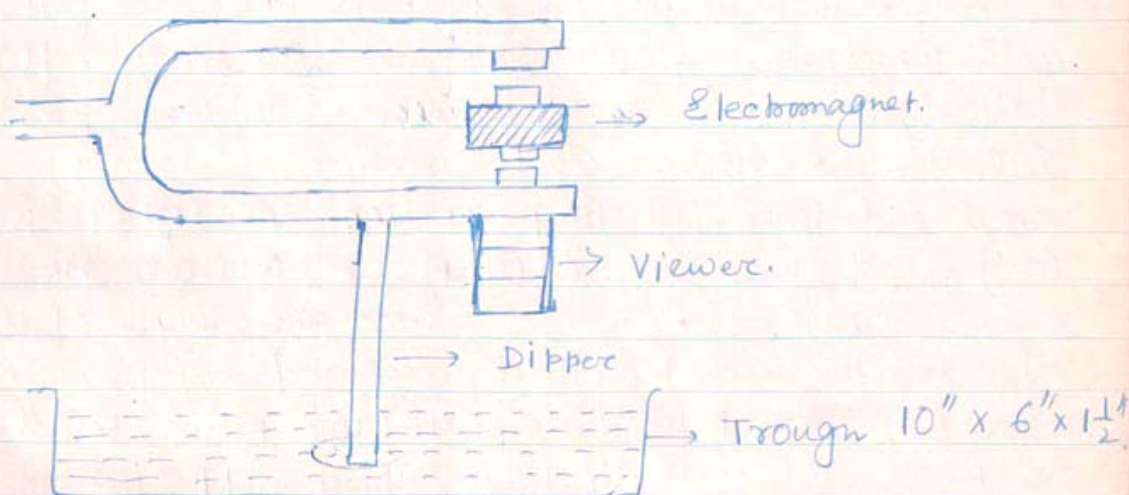
gravitational force per unit mass hence following the analogy the term $\frac{4\pi^2 s}{\rho\lambda^2}$ can be called



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Case III : For $\lambda = 1.7 \text{ cm}$ from Ratio ≈ 1 i.e. both surface tension and gravity are equally important. Hence for experimental determination of S.T of a liquid, by this method we must prefer waves having wave length around 1.7 cm .

Experimental determination of Surface Tension of water by Ripple-Tank method :-



1. It consists of an electrically maintained tuning fork, which vibrates with a frequency about 100 per sec.
2. Water is taken in a trough of size $10'' \times 6'' \times 1\frac{1}{2}''$
3. A dipper which is a blade of highly polished aluminium or silver $3''$ in length attached with the lower prong, with its plane vertical and perpendicular to the plane of vibration of the prong. The dipper moves up and down in the liquid and produces waves. The 1st



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wave after reflection from the wall of the trough superpose with the second wave, travelling in opposite direction and gives rise to stationary wave.

4) A wire net known as viewer; which is a rectangular frame; which have two steel wires attached on it is fixed on the lower prong at the edge with its plane vertical and parallel to the plane of vibration of the fork and observations are made through the viewer.

5) A parallel beam of light is made to fall vertically on the liquid surface which are reflected obliquely from the wavy liquid surface, pass through the viewer and produce a stationary image of the vibrating viewer. The image is wavy and we assume that the wave length of the image is same as the wave length of the ripples. The wave length is measured by photographic method. A metal plate with two fine notches (marks) on it at a fixed distance apart helps the measure.

6) The frequency of vibration of the tuning fork is measured by a stroboscopic disc which is turned in unison with a standard flickering neon lamp.

Thus knowing n , λ , ρ ; S can be calculated using equation (13):

$$S = \frac{n^2 \lambda^3 \rho}{2\pi} - \frac{g \rho \lambda^2}{4\pi}$$